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31/07/2024

Code-A Phase-1

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 | NCERT Booster Test Series for NEET (RM)-2024-25_T01A | Time : 200 Min.

Topics Covered:
Physics: Units & Measurements, Motion in a Straight Line, Motion in a Plane, Laws of Motion, Work, Energy & Power, System of Particles & Rotational Motion
Chemistry: Some Basic Concepts of Chemistry, Structure of Atom, Classification of Elements and Periodicity in Properties, Chemical Bonding and Molecular Structure
Botany: Cell : The Unit of Life, Cell Cycle & Cell Division, The Living World, Biological Classification
Zoology: Structural Organisation in Animals– Animal Tissues, Biomolecules, Breathing & Exchange of Gases, Body Fluids & Circulation

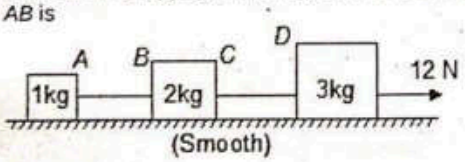
General Instructions :

1. There are two sections in each subject, i.e. **Section-A & Section-B**. You have to attempt all **35 questions** from Section-A & only **10 questions** from Section-B out of **15.**
2. Each question carries **+4 marks**. For every wrong response, **–1 mark** shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
3. Use blue/black ballpoint pen only to darken the appropriate circle.
4. Mark should be dark and completely fill the circle.
5. Dark only one circle for each entry.
6. Dark the circle in the space provided only.
7. Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

PHYSICS

SECTION-A

1. All the strings shown in figure are massless. If horizontal force 12 N is applied on 3 kg, then tension in the string *AB* is



(1) 2 N
(2) 3 N
(3) 6 N
(4) 12 N

2. A boy of mass 40 kg wants to climb up a rope hanging vertically. The rope can withstand a maximum tension of 600 N. The maximum acceleration with which the boy can climb the rope is (g = 10 m/s^2)

(1) 10 m/s^2
(2) 5 m/s^2
(3) 2.5 m/s^2
(4) 7.5 m/s^2

3. A ball is thrown vertically upward with velocity u from the top of a building reaches the ground with a speed $3u$. The height of tower is

(1) $3\frac{u^2}{g}$

(2) $2\frac{u^2}{g}$

(3) $4\frac{u^2}{g}$

(4) $\frac{u^2}{2g}$

4. A particle starts moving with constant acceleration 3 m/s^2. Distance travelled by the particle in 4th second of its journey is

(1) 24 m

(2) 37.5 m

(3) 10.5 m

(4) 15 m

5. A body of mass m is projected with initial speed u at an angle θ with horizontal. The magnitude of change in momentum after the body reaches its maximum height is

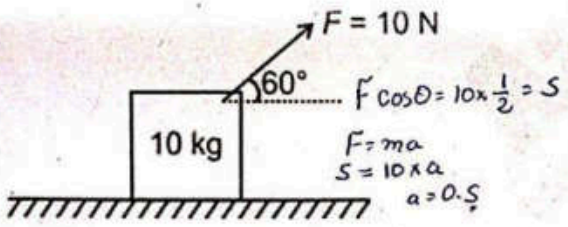
(1) $2mu\cos\theta$

(2) Zero

(3) $mu\sin\theta$

(4) $2mu\sin\theta$

6. The acceleration of block A (of mass 10 kg), kept on a smooth horizontal surface, when a force F of magnitude 10 N is applied on it as shown in figure will be



(1) 0.25 m/s^2

(2) 2 m/s^2

(3) 1 m/s^2

(4) 0.5 m/s^2

7. What will be the value of $\frac{H^2}{T^4}$ in oblique projection from ground to ground projection? (H = Maximum height T = Time of flight)

(1) 5 : 4

(2) 4 : 5

(3) 25 : 16

(4) 16 : 25

8. The moment of inertia about hinge axis of a wheel of radius 20 cm is 40 kg m^2. If a tangential force of 100 N is applied on rim of the wheel. Then its angular acceleration will be

(1) 1 rad/s^2

(2) $\frac{1}{2}$ rad/s^2

(3) 2 rad/s^2

(4) 4 rad/s^2

9. A particle is released from rest at origin. It moves under the influence of a potential field $U = (x^2 - 3x)$ J. Its kinetic energy at $x = 2$ is

(1) 2 J

(2) 1 J

(3) Zero

(4) $\frac{3}{2}$ J

10. A : In every collision, linear momentum of the system is conserved.
R : In every collision, kinetic energy of the system is conserved.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

11. The percentage error in the measurement of mass and speed are 3% and 4% respectively. Maximum percentage error in kinetic energy is (Take K.E. = $\frac{1}{2}mv^2$)

(1) 8%

(2) 7%

(3) 10%

(4) 11%

12. A : If momentum of a body increases by 10%, its kinetic energy will increase by 21%.
R : Kinetic energy of a given body is proportional to square of momentum.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

13. The position x (in m) of a particle is given by $3x = a + bt^3$, where a and b are constants. Acceleration of particle at t = 2 s is

(1) $\frac{4b}{3}$ m/s^2

(2) $\frac{2b}{3}$ m/s^2

(3) $2b$ m/s^2

(4) $4b$ m/s^2

14. **Statement A:** Energy is cause for doing work and work is the effect.
Statement B: Energy is scalar quantity but power is a vector quantity.

(1) A and B are correct

(2) A and B are incorrect

(3) A is correct and B is incorrect

(4) A is incorrect and B is correct

15. A wheel initially at rest, starts rotating with a uniform angular acceleration. The wheel rotates through an angle θ_1 in first second and through an additional angle θ_2 in next two second. The value of $\frac{\theta_2}{\theta_1}$ is

(1) 8

(2) 4

(3) $\frac{1}{4}$

(4) $\frac{1}{8}$

16. A particle has an initial velocity $\left(4\hat{i}+2\hat{j}\right)$ and an acceleration of $\left(0.8\hat{i}+0.3\hat{j}\right)$. Its speed after 10 s is

(1) 5 units

(2) 7 units

(3) 13 units

(4) $2\sqrt{3}$ units

17. A: If velocity and acceleration of a particle are always perpendicular to each other, then speed decreases.
R: Kinetic energy in this case will decrease.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

18. Acceleration of a particle varies with time as $a = 12t^2$ m/s^2. If the particle starts its motion from rest, then its displacement in initial 2 seconds will be

(1) 16 m

(2) 24 m

(3) 64 m

(4) 4 m

19. If time (T), force (F) and velocity (V) are chosen as fundamental quantities then dimensional formula of power is

(1) $[FVT^0]$

(2) $[FVT]$

(3) $[FV^2T]$

(4) $[FVT^{-1}]$

20. A : Power developed in uniform circular motion is always zero.
R : Kinetic energy in uniform circular motion is variable.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

21. A fan is rotating with a speed of 360 rev/minute. It comes to rest in 6 s, on switching off the switch. The number of revolutions made by it before coming to rest is

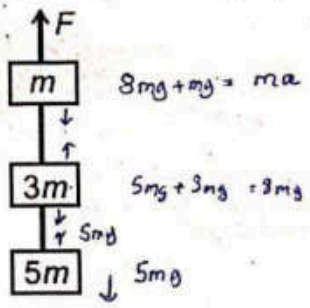
(1) 9

(2) 27

(3) 18

(4) 36

22. Three blocks with masses m, $3m$, $5m$ are connected by strings as shown in figure. An upward force F is applied on block of mass m, the masses moves upwards with constant speed v. Then the net force acting on mass $3m$ is (acceleration due to gravity is g)



(1) 2 mg

(2) 3 mg

(3) 5 mg

(4) Zero

23. If force $F = (x^2 - 3x + 2)$ N acting on particle, then stable equilibrium of particle is at (Where x is in meter and F is in newton)

(1) $x = -1$ m

(2) $x = 1$ m

(3) $x = 2$ m

(4) $x = -2$ m

24. Position of a particle can be expressed as $\vec{r} = \left(9t^2\hat{i} + 3t\hat{j}\right)$ m. The magnitude of acceleration of the particle is (t is in second)

(1) 18 m/s^2

(2) 9 m/s^2

(3) Zero

(4) $9\sqrt{5}\,m/s^2$

25. The linear density of a thin rod of length 1 m varies as $\lambda = 3x^2 + 2$, where x is the distance from its one end. The distance of centre of mass from this end is

(1) $\frac{1}{2}m$

(2) $\frac{3}{4}m$

(3) $\frac{5}{12}$ m

(4) $\frac{7}{12}$ m

26. A spring of force constant 500 N/m has an extension of 4 cm. The work done in extending it from 4 cm to 12 cm is

(1) 3.2 J

(2) 1.6 J

(3) 0.8 J

(4) 6.4 J

27. Rain is falling vertically downwards. For a man walking at a speed 6 m/s on ground, the rain appears to fall at an angle 45° with vertical, towards the man. The speed of the rain w.r.t. ground is

(1) 2 m/s

(2) 3 m/s

(3) $2\sqrt{2}$ m/s

(4) 6 m/s

28. In vertical circular motion if the speed of particle of mass 1 kg at lowest point is 20 m/s, then the tension at highest point is (Take radius of circular loop is 2 m)

(1) 100 N

(2) 150 N

(3) 50 N

(4) Zero

29. A body whose momentum is non zero constant, must have constant

(1) Non-zero force

(2) Non-zero velocity

(3) Non-zero acceleration

(4) All of these

30. If 25th division of Vernier scale coincides with 20th division of main scale, then the least count of Vernier Calliper is (Take 1 MSD = 1 mm)

(1) 0.1 mm

(2) 0.2 mm

(3) 0.01 mm

(4) 0.02 mm

31. A person travels along a straight road for first $\frac{t}{5}$ time with speed v_1 and for next $\frac{4t}{5}$ time with a speed v_2. The mean speed v is given by

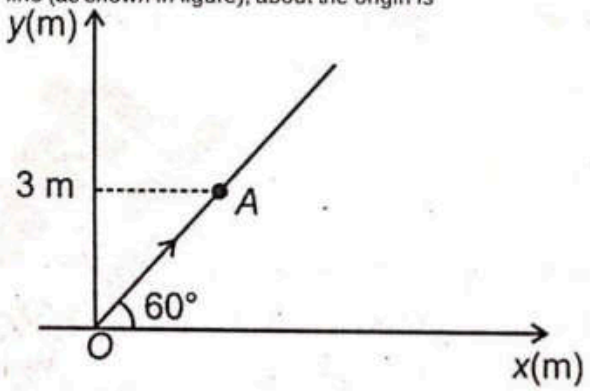
(1) $v = \frac{v_1 + v_2}{2}$

(2) $v = \frac{v_1 + 4v_2}{5}$

(3) $\frac{1}{v} = \frac{1}{5v_1} + \frac{4}{5v_2}$

(4) $v = \frac{2}{5}\sqrt{v_1 v_2}$

32. The angular momentum [in kg-m^{2}s^{-1}] of a particle of mass 5 kg, moving with a speed of 20 m/s in a straight line (as shown in figure), about the origin is



(1) $100\sqrt{2}$

(2) Zero

(3) 300

(4) $\frac{300}{\sqrt{2}}$

33. A disc rotates about its axis from rest, acquires the angular speed 40 rev/s in 4 seconds with constant angular acceleration. The angle rotated by it (in radians) during t = 2 s to t = 4 s is

(1) 30π

(2) 60π

(3) 120π

(4) 180π

34. A sphere is rotating about its diameter with uniform angular velocity, then

(1) Acceleration of particles on surface of sphere is zero

(2) Acceleration of particles at a distance half of radius from axis is zero

(3) All particles have same linear speed

(4) Particles may have different linear speeds

35. A missile is fired for maximum range with an initial velocity 40 m/s. The range of the missile is (g = 10 m/s^2)

(1) 60 m

(2) 160 m

(3) 200 m

(4) 40 m

SECTION-B

36. A : Friction always decreases the speed of body.
R : Friction always opposes the motion of body.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

37. A park in the form of a square has an area of (100 ± 0.4) m^2. The side of park is

(1) (10 ± 0.02) m

(2) (10 ± 0.03) m

(3) (10 ± 0.04) m

(4) (10 ± 0.3) m

38. A: The time spent in river will be minimum when swimmer swims in the direction perpendicular to the velocity of river.
R: In this case distance travelled by the swimmer along the river bank will also be minimum.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

39. The motion of a particle along a straight line is described by equation $x = 2 + 12t - t^2$, where x is in m and t is in s. The retardation of the particle when its velocity becomes zero is

(1) 2 m/s^2

(2) 3 m/s^2

(3) 4 m/s^2

(4) 5 m/s^2

40. In equation $y = A\cos^2(\omega t - kx)$, where t is time and x is distance, dimensional formula of ω/k should be

(1) $[LT^{-1}]$

(2) $[L^{-1}T]$

(3) $[L^1T^1]$

(4) $[L^0T^0]$

41. A wheel having moment of inertia 4 kg m^2 about its symmetrical axis, rotates at rate of 240 rpm about it. The torque which can stop the rotation of the wheel in one minute is

(1) $\frac{3\pi}{7}$ N m

(2) $\frac{2\pi}{9}$ N m

(3) $\frac{8\pi}{15}$ N m

(4) $\frac{5\pi}{7}$ N m

42. A machine gun fires 5 bullets per second and the mass of each bullet is 20 g. If 100 m/s is the speed of each bullet, then the force exerted on the machine gun is

(1) 10 N

(2) 20 N

(3) 8 N

(4) 15 N

43. A uniform rod is 4 m long and weight is 10 kg. If it is supported on a knife edge at one metre from one end, what weight should be placed at that end to keep the rod horizontal?

(1) 8 kg

(2) 10 kg

(3) 12 kg

(4) 16 kg

44. The position of a particle along *x*-axis at time *t* is given by $x = 1 + t - t^2$. The distance travelled by the particle in first 2 seconds is

(1) 1 m

(2) 2 m

(3) 2.5 m

(4) 3 m

45. A circular disc of mass 5 kg and radius 1 m rotating with speed 60 rpm about an axis passing through a point on its circumference and lies in the plane of the disc. The kinetic energy of the disc will be nearly

(1) 250 J

(2) 125 J

(3) 110 J

(4) 220 J

46. If a boy standing on a stationary boat in a pond, moves on the boat to his right, then the

(1) Boat also moves to his right

(2) Boat remains stationary

(3) Centre of mass of the system (boat + boy) moves to his left

(4) Centre of mass of the system (boat + boy) remains stationary

47. A vector perpendicular to both vector $\vec{A} = \hat{i} + 2\hat{j} + \hat{k}$ as well as $\vec{B} = \hat{i} + \hat{j} - \hat{k}$ is

(1) $3\hat{i} + 2\hat{j} - \hat{k}$

(2) $-6\hat{i} + 4\hat{j} - 2\hat{k}$

(3) $-6\hat{i} + 4\hat{j} + 2\hat{k}$

(4) $3\hat{i} + 2\hat{j} + \hat{k}$

48. If $\vec{A} = 4\hat{i} + 4\hat{j}$, After rotating $\vec{A}$ by 45° anticlockwise then new vector obtained is

(1) $4\hat{i}$

(2) $4\hat{j}$

(3) $4\sqrt{2}\hat{i}$

(4) $4\sqrt{2}\hat{j}$

49. A force $F(t) = (6 - 8t)$ N varying with time (t) acts on a particle moving along x-axis. The magnitude of impulse imparted to the particle from $t = 0$ to $t = 10$ s is

(1) 300 Ns

(2) 340 Ns

(3) 210 Ns

(4) 120 Ns

50. The elastic potential energy stored in a spring, having a compression 10 cm, is 20 J. The potential energy stored in spring, when it is given an extension of 10 cm from its natural length, is

(1) 40 J

(2) 20 J

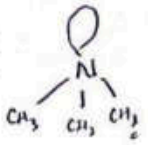
(3) 10 J

(4) 5 J

CHEMISTRY

SECTION-A

51. The shapes of the molecules $(CH_3)_3N$ and $(SiH_3)_3N$ respectively are



(1) Trigonal planar and pyramidal

(2) Pyramidal and trigonal planar

(3) Both pyramidal

(4) Both trigonal planar

52. 3.01×10^{22} molecules of urea is present in 100 ml of an aqueous solutions. The molarity of the solution is

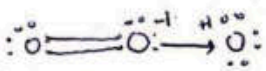
(1) 0.05 M

(2) 2.5 M

(3) 0.5 M

(4) 1.5 M

53. The formal charge on the central oxygen atom in O_3 molecule is



(1) +1

(2) −1

(3) 0

(4) +2

54. Which of the following transitions in hydrogen spectrum produces spectral line in UV region?

(1) 4 → 2

(2) 5 → 3

(3) 3 → 1

(4) 7 → 3

55. Atomic number of element whose symbol is Unb is

(1) 102

(2) 103

(3) 104

(4) 101

56. The de-Broglie wavelength associated with a particle of mass 10 mg moving with velocity of 10 ms^{-1}, is

(1) 6.62×10^{-36} m

(2) 6.62×10^{-30} m

(3) 6.62×10^{-33} m

(4) 6.62×10^{-32} m

57. Total number of electrons present in 12 g of Mg^{2+} ion is

(1) 3.01×10^{24}

(2) 6.02×10^{23}

(3) 3.01×10^{23}

(4) 6.02×10^{24}

58. Correct order of electron affinity of given elements is

(1) F > Cl > Br > I

(2) I > Br > Cl > F

(3) Cl > F > Br > I

(4) I > Br > F > Cl

59. Which of the following occupies maximum volume at STP?

(1) 0.1 mole $CO_2(g)$

(2) 11 g $CO_2(g)$

(3) 3×10^{20} molecules of $CO_2(g)$

(4) 1 L of $CO_2(g)$

60. Identify the incorrect match from the options.

	Name		IUPAC Official Name
a.	Unnilbium	(i)	Nobelium
b.	Unnilquadium	(ii)	Rutherfordium
c.	Unnilseptium	(iii)	Bohrium
d.	Unnilennium	(iv)	Dubnium

(1) a, (i)

(2) b, (ii)

(3) c, (iii)

(4) d, (iv)

61. If the mass ratio of helium and methane is 1 : 8 then the ratio of the moles of two gases will be

(1) 1 : 8

(2) 1 : 4

(3) 1 : 1

(4) 1 : 2

62. If the energy of third orbit of He^+ ion is $-x$ kJ mol^{-1}, then the energy of second orbit of H-atom (in kJ mol^{-1}) will be

(1) $-\frac{9x}{16}$

(2) $-\frac{3x}{4}$

(3) $-\frac{9x}{4}$

(4) $-\frac{3x}{16}$

63. Wavelength of the electron in the third orbit of hydrogen atom is (Radius of Bohr's first orbit of hydrogen is 52.9 pm)

(1) 215.2 π pm

(2) 317.4 π pm

(3) 412.2 π pm

(4) 125.6 π pm

64. Shape of XeO_2F_2 is

(1) Tetrahedral

(2) Square pyramidal

(3) See-saw

(4) T-shaped

65. If uncertainty in position and momentum of a moving particle is equal then uncertainty in its velocity will be

(1) $\frac{1}{m}\sqrt{\frac{h}{\pi}}$

(2) $\frac{1}{3m}\sqrt{\frac{h}{\pi}}$

(3) $\frac{1}{4m}\sqrt{\frac{h}{\pi}}$

(4) $\frac{1}{2m}\sqrt{\frac{h}{\pi}}$

66. What is the percentage purity of magnesium carbonate sample whose 8.4 g sample on decomposition gives 2.2 g CO_2?

(1) 25%

(2) 100%

(3) 50%

(4) 75%

67. Which of the following oxides is neutral?

(1) CaO

(2) N_2O

(3) SO_3

(4) Cl_2O_7

68. A neutral atom Ar is converted to Ar^{3+} by the following process :

$Ar(g) \xrightarrow[-e]{E_1} Ar^{+}(g) \xrightarrow[-e]{E_2} Ar^{2+}(g) \xrightarrow[-e]{E_3} Ar^{3+}(g)$

The correct order of E_1, E_2 and E_3 energies are

(1) $E_1 < E_2 < E_3$

(2) $E_1 > E_2 > E_3$

(3) $E_1 = E_2 = E_3$

(4) $E_1 > E_2 < E_3$

69. Match the following items in column I with column II and choose the correct option.

	Column I (quantum no. s)		Column II (orbital)
(a)	n = 2, l = 1	(i)	3p
(b)	n = 3, l = 2	(ii)	3d
(c)	n = 1, l = 0	(iii)	2p
(d)	n = 3, l = 1	(iv)	1s

(1) (a)(iii), (b)(i), (c)(iv), (d)(ii)

(2) (a)(iii), (b)(ii), (c)(iv), (d)(i)

(3) (a)(ii), (b)(i), (c)(iii), (d)(iv)

(4) (a)(i), (b)(ii), (c)(iii), (d)(iv)

70. Volume of O_2 required for the complete combustion of 11.2 L of CH_4 at STP is

(1) 11.2 L

(2) 22.4 L

(3) 33.6 L

(4) 44.8 L

71. Maximum number of lone pairs of electrons are present in

(1) XeO_2F_2

(2) IF_3

(3) BF_4^-

(4) XeF_4

72. An element (X) has equal number of neutrons as are in $_{16}S^{32}$ and has 31 nucleons. The position of element (X) in periodic table is

(1) 3rd period, 16th group

(2) 4th period, 14th group

(3) 3rd period, 15th group

(4) 4th period, 15th group

73. Which is the incorrect statement related to cathode rays?

(1) These rays move from anode to cathode

(2) These rays consist of negatively charged particles called electrons

(3) These rays do not depend upon the nature of the gas present in the cathode ray tube

(4) These rays travel in straight line in the absence of electric and magnetic field

74. Mass percentage of iron in a biomolecule is 0.1%. The minimum molecular mass of the biomolecule is (Atomic mass of Fe = 56 u)

(1) 5600 g

(2) 5.6×10^5 g

(3) 56000 u

(4) 5600 u

75. The dipole moment of HBr is 1.6×10^{-30} coulomb metre and interatomic distance is 1 Å. The percentage ionic character of HBr is

(1) 10

(2) 7

(3) 15

(4) 26

76. Correct order of wavelength of the given radiations is

(1) Radiowaves > IR > Microwave

(2) IR > Microwave > Radiowaves

(3) Radiowaves > Microwave > IR

(4) Microwave > Radiowaves > IR

77. Correct order of stability is

(1) $N_2^+ > N_2^- > N_2$

(2) $N_2 > N_2^- > N_2^+$

(3) $N_2 > N_2^+ > N_2^-$

(4) $N_2^- > N_2 > N_2^+$

78. The ratio of the wavelength of the first line of Paschen series to that of the second line of Balmer series of H atom is

(1) 25 : 8

(2) 8 : 25

(3) 27 : 7

(4) 7 : 27

79. The correct set of quantum numbers for the valence shell electron of Ca (Atomic number : 20) is

(1) $n = 4, l = 0, m = 0, s = -\frac{1}{2}$

(2) $n = 3, l = 0, m = 0, s = +\frac{1}{2}$

(3) $n = 3, l = 1, m = 0, s = +\frac{1}{2}$

(4) $n = 3, l = 0, m = 0, s = -\frac{1}{2}$

80. An element X has the following isotopic composition.
^{100}X : 80%, ^{102}X : 20%.
The weighted average atomic mass of the naturally occurring element X is

(1) 101.1 u

(2) 100.4 u

(3) 100.8 u

(4) 101.8 u

81. Find the **correct** match.

	Column I		Column II
a.	BCl_3	(i)	sp^2
b.	$SiCl_4$	(ii)	sp^3d
c.	SF_4	(iii)	sp^3d^2
d.	BrF_5	(iv)	sp^3

(1) a(i), b(ii), c(iii), d(iv)

(2) a(i), b(iv), c(ii), d(iii)

(3) a(i), b(ii), c(iv), d(iii)

(4) a(iv), b(i), c(ii), d(iii)

82. Correct order of penetrating power of α-particles, β-rays and γ-rays is

(1) γ-rays > β-rays > α-particles

(2) α-particles > β-rays > γ-rays

(3) γ-rays > α-particles > β-rays

(4) β-rays > α-particles > γ-rays

83. Angular momentum of electron in the third orbit of hydrogen atom is

(1) $\frac{h}{2\pi}$

(2) $\frac{2h}{3\pi}$

(3) $\frac{3h}{2\pi}$

(4) $\frac{h}{3\pi}$

84. The ionic radii of N^{3-}, O^{2-}, F^-, Na^+ follow the order

(1) $N^{3-} > O^{2-} > F^- > Na^+$

(2) $N^{3-} > Na^+ > O^{2-} > F^-$

(3) $Na^+ > O^{2-} > N^{3-} > F^-$

(4) $Na^+ > F^- > O^{2-} > N^{3-}$

85. Consider the following observations for CO_2 sample

	% of carbon	% of oxygen
Natural sample	27.27	72.73
Synthetic sample	27.27	72.73

The above observation is in accordance to

(1) Law of multiple proportions

(2) Law of definite proportions

(3) Law of conservation of mass

(4) Gay-Lussac's law of gaseous volume

SECTION-B

86. Find number of oxygen atoms present in 100 mg of $CaCO_3$.
(Atomic mass of Ca = 40 u, C = 12 u, O = 16 u)

(1) 6.02×10^{23}

(2) 6.02×10^{20}

(3) 1.806×10^{21}

(4) 1.204×10^{20}

87. Given below are two statements:
Statement-I: The magnitude of H-bonding of a compound is maximum in its solid state.
Statement-II: HCl does not show H-bonding with its molecules.
In the light of above statements, choose the correct answer from the options given below.

(1) Statement I is true but statement II is false

(2) Statement I is false but statement II is true

(3) Both statements I and II are true

(4) Both statements I and II are false

88. Mole fraction of solute in a 4.45 molal aqueous solution is

(1) 0.121

(2) 0.021

(3) 0.074

(4) 0.205

89. Maximum number of atoms present in a plane in SbF_5 is

(1) 5

(2) 4

(3) 3

(4) 2

90. Energy of one mole of photons of radiation whose frequency is 10^{14} Hz is ($h = 6 \times 10^{-34}$ Js) ($N_A = 6 \times 10^{23}$)

(1) 3.6×10^{-9} J

(2) 3.6×10^{-19} J

(3) 3.6×10^{4} J

(4) 3.6×10^{14} J

91. Number of Cl = O bond(s) in $HClO_4$ molecule is

(1) 1

(2) 2

(3) 3

(4) 4

92. Consider the following statements
Statement I: Total number of electrons present in π antibonding molecular orbital of N_2 is zero.
Statement II: Total number of antibonding electrons in N_2 is 4.
In light with above statements choose the correct option.

(1) Both the statements I and II are correct

(2) Both the statements I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

93. An oxide of nitrogen contains 69.5% oxygen, empirical formula of oxide is

(1) N_2O

(2) N_2O_3

(3) NO_2

(4) N_2O_5

94. Al^{3+} ion is isoelectronic with

(1) O^{2-}

(2) Ca^{2+}

(3) Cl^-

(4) K^+

95. Select the molecule which does not contain any π bond.

(1) SO_2

(2) CO_2

(3) ICl_3

(4) N_2

96. Which of the following pairs of d-orbitals will have electron density along the axes?

(1) $d_{x^2-y^2}, d_{z^2}$

(2) d_{xy}, d_{z^2}

(3) $d_{yz}, d_{x^2-y^2}$

(4) d_{xy}, d_{yz}

97. 28 kg of $N_2(g)$ and 3 kg of $H_2(g)$ are mixed to produce $NH_3(g)$. The maximum amount of $NH_3(g)$ formed is

(1) 34 kg

(2) 17 kg

(3) 28 kg

(4) 8.5 kg

98. Electronic configuration of an element (M) is $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^1$. Formula of its oxide is

(1) MO

(2) M_2O

(3) M_2O_3

(4) MO_2

99. During change of O_2 to O_2^- ion, the electron adds to which one of the following molecular orbitals?

(1) π^*

(2) π

(3) σ^*

(4) σ

100. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**.

A : ψ^2 at a point gives the probability density of the electron at that point.

R : For 1 s orbital the probability density is minimum at the nucleus.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

BOTANY

SECTION-A

101. Lysosomes

(1) Are membraneless

(2) Are formed by the process of packaging in Golgi apparatus

(3) Contain enzymes active at basic pH

(4) Are not part of endomembrane system

102. Which of the following is **not** the function of mesosome in bacteria?

(1) DNA replication

(2) Cell wall formation

(3) Photosynthesis

(4) Respiration

103. Fat soluble carotenoids pigments like carotene and xanthophylls are mainly present in

(1) Amyloplast

(2) Aleuroplast

(3) Chromoplast

(4) Elaioplast

104. Select the **incorrectly** matched pair.

(1)	Bacterial flagellum	–	Filament, Hook and basal body
(2)	Primary cell wall in plants	–	Incapable of growth
(3)	White blood cells	–	Amoeboid
(4)	Ribosomes	–	Non-membrane bound organelles

(1) (1)

(2) (2)

(3) (3)

(4) (4)

105. Read the following statements and choose the **correct** option.

Assertion: The content of nucleolus is continuous with the rest of the nucleoplasm.

Reason: Nucleolus is not a membrane bound structure.

(1) Both Assertion and Reason are true and Reason is the correct explanation for Assertion

(2) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion

(3) Only Assertion is true

(4) Only Reason is true

106. What is the main arena of all cellular activities of a cell?

(1) Cell wall

(2) Cell membrane

(3) Nucleus

(4) Cytoplasm

107. Thick and tough glycocalyx found in some bacteria is called

(1) Slime layer

(2) Mesosome

(3) Capsule

(4) Lamellae

108. The core of eukaryotic cilia and flagella is called axoneme which is made up of

(1) Filament, hook and basal body

(2) 9 centrally located doublets and 2 microtubules at periphery

(3) 2 centrally located microtubules and 9 doublets at periphery

(4) Central hub and 11 peripheral doublets

109. Which of the following can be the site of ATP synthesis in plant cells?

a. Nucleoplasm
b. Golgi apparatus
c. Chloroplast
d. Mitochondrion

Select the correct option.

(1) (a) and (d) only

(2) (c) and (d)

(3) (b) and (d) only

(4) (c) only

110. Which of the following statements is **incorrect** w.r.t. Endoplasmic Reticulum.

(1) RER is actively involved in protein synthesis and secretion

(2) SER is the major site for synthesis of lipids

(3) It divides the intracellular space into three distinct compartments

(4) In animal cells lipid like steroidal hormones are synthesised in SER

111. All of the following phenomena may occur twice in meiosis, **except**

(1) Spindle formation

(2) Uncoiling of chromosomes

(3) DNA duplication

(4) Centriole duplication

112. The X-shaped structures formed by the homologous chromosomes in diplotene stage are

(1) Chiasmata

(2) Kinetochores

(3) Recombination nodule

(4) Telomeres

113. In animal cell, the phase between G_1 and G_2 phase can be characterised by duplication of

(1) Centriole

(2) Mitochondria

(3) Golgi body

(4) Chloroplast

114. Pairing of homologous chromosomes called synapsis takes place in which phase of prophase I?

(1) Zygotene

(2) Diplotene

(3) Pachytene

(4) Leptotene

115. Crossing over takes place between

(1) Sister chromatids of a chromosome

(2) Non-sister chromatids of homologous chromosomes

(3) Sister chromatids of non homologous chromosomes

(4) Non-sister chromatids of non homologous chromosomes

116. Anaphase I differs from anaphase of mitosis as in the former

(1) Separation of homologous chromosomes does not occur

(2) Splitting of centromere occurs

(3) Sister chromatids do not separate

(4) Chromosomes cluster at the equator

117. A bivalent of prophase I represents

(1) One centromere with two chromatids

(2) Two centromeres with two chromatids

(3) Two centromere with four chromatids

(4) Four centromere with four chromatids

118. The daughter cells formed after meiosis-II are

(1) Similar to each other but dissimilar to parent cell

(2) Similar to parent cell but dissimilar to each other

(3) Similar to each other as well as to parent cell

(4) Neither similar to each other nor to parent cell

119. Which of the following stages can last for months or years in oocytes of some vertebrates?

(1) Diakinesis

(2) Diplotene

(3) Pachytene

(4) Anaphase I

120. The Muscidae of housefly is equivalent to _____ of mango. Complete the statement by selecting **correct** option to fill up the blank.
(1) Sapindales
(2) Anacardiaceae
(3) Dicotyledonae
(4) Poales

121. Plant families like Convolvulaceae and Solanaceae are included in the order Polymoniales mainly based on
(1) Habitat and distribution
(2) Floral characters
(3) Vegetative characters
(4) Root and stem structures

122. Out of the seven taxonomic categories, the basic and the lowest category of taxonomic hierarchy is
(1) Species
(2) Genus
(3) Tribe
(4) Family

123. Systematics differs from taxonomy as the former includes
(1) Identification of organisms
(2) Characterisation of organisms
(3) Nomenclature of organisms
(4) Evolutionary relationship among organisms

124. Fruiting body of *Aspergillus* is called
(1) Conidium
(2) Basidiocarp
(3) Ascocarp
(4) Sporangium

125. Select the **incorrect** statement about dinoflagellates.
(1) They are eukaryotes
(2) The cell wall has stiff cellulosic plates
(3) *Gonyaulax* is a red dinoflagellate
(4) Most of them have single flagellum

126. Syncytium formation is due to
(1) Failure of karyokinesis after cytokinesis
(2) Failure of cytokinesis after karyokinesis
(3) Failure of both karyokinesis and cytokinesis
(4) Occurrence of cytokinesis after karyokinesis

127. In *Agaricus*, karyogamy and meiosis take place in
(1) Ascus
(2) Basidium
(3) Zygospore
(4) Basidiospore

128. Red tide in sea is caused by
(1) *Anabaena*
(2) *Gonyaulax*
(3) Diatoms
(4) *Entamoeba*

129. _____ is used extensively in biochemical and genetic work.
Select the **correct** option to fill in the blank.
(1) *Neurospora*
(2) *Claviceps*
(3) *Aspergillus*
(4) *Agaricus*

130. The vast majority of bacteria are
(1) Chemosynthetic autotrophs
(2) Heterotrophs
(3) Photoautotrophs
(4) Capable of synthesizing their own food

131. Members of deuteromycetes differ from the members of ascomycetes as the former
(1) Have aseptate mycelium
(2) Have endogenously produced asexual spores
(3) Lack sexual spores
(4) Lack parasitic members

132. Which one is **incorrect** w.r.t. lichen?
(1) Lichens are symbiotic association between algae and fungi
(2) Algal component is known as phycobiont
(3) Algae prepare food and fungi in return provide shelter, absorb mineral nutrients and water for its partner
(4) Lichens are very good pollution indicators as they grow well in polluted areas

133. The protozoan which causes sleeping sickness is a/an
(1) Sporozoan
(2) Ciliated protozoan
(3) Flagellated protozoan
(4) Amoeboid protozoan

134. Chemosynthetic autotrophic bacteria
(1) Oxidise various inorganic substances
(2) Use light energy for their ATP production
(3) Play no role in recycling of nutrients
(4) Often form blooms in polluted water bodies

135. All of the following are order, **except**
(1) Primata
(2) Carnivora
(3) Mammalia
(4) Diptera

SECTION-B

136. Many ribosomes may associate with a single mRNA. This structure is called

(1) Polysome

(2) Nucleosome

(3) Chromosome

(4) Glyoxysome

137. Bacteria are classified into Gram positive and Gram negative on the basis of differences in the

(1) Cell envelope and response to a particular stain

(2) Elements present in the cell membrane

(3) Sequence of 16S ribosomal RNA genes

(4) Length of the plasmid DNA

138. The membrane proteins are classified as integral and peripheral. The basis of this classification is

(1) Quasi-fluid nature

(2) Biochemical investigation

(3) Ease of extraction

(4) Requirement of energy for transport of material

139. How many of the following is/are the function(s) of cell wall?

(a) Gives shape to the cells

(b) Protects from mechanical damage and infection

(c) Regulates the transport of the molecules across it

(d) Helps in cell to cell interaction

(1) One

(2) Two

(3) Three

(4) Four

140. A cell at prophase shows all, **except**

(1) Condensation of chromatin

(2) Spindle fibre formation

(3) Nucleolus reappearance

(4) Disintegration of nuclear membrane

141. Chromosomes are thickest and shortest and their morphology is easily studied in

(1) Anaphase

(2) Metaphase

(3) Prophase

(4) Telophase

142. Interkinesis is a metabolic stage between a and b.

	(a)	(b)
(1)	Telophase II	Prophase I
(2)	Telophase II	Prophase II
(3)	Telophase I	Prophase I
(4)	Telophase I	Prophase II

(1) (1)

(2) (2)

(3) (3)

(4) (4)

143. In the resting phase of cell cycle

(1) There is no metabolic process in the cell

(2) Only DNA replication occurs in the nucleus

(3) Cell growth and DNA replication occur in orderly manner

(4) Only cytoplasmic content becomes double

144. Scientific name of mango is correctly printed as

(1) Mangifera indica

(2) Mangifera Indica

(3) *Mangifera Indica*

(4) *Mangifera indica*

145. Select the option showing hierarchial arrangement of taxonomic categories in ascending order

(1) Species → Genus → Family → Order

(2) Genus → Species → Family → Order

(3) Genus → Species → Order → Family

(4) Species → Family → Order → Genus

146. Which of the following taxa belong to the same rank?

(1) *Homo*, *Felis*, *pardus*

(2) Solanaceae, *leo*, Convolvulaceae

(3) *Mangifera*, *Solanum*, *Panthera*

(4) Hominidae, *Triticum*, Dicotyledonae

147. Causal agent of bovine spongiform encephalopathy

(1) Contains RNA as genetic material

(2) Is devoid of proteins

(3) Is abnormally folded protein

(4) Contains dsDNA

148. Select the **incorrect** match from the following.

(1) *Albugo* – Zoospore formation

(2) *Rhizopus* – Aseptate and coenocytic mycelium

(3) Mycorrhiza – Symbiotic association

(4) *Rhizopus* – Produce conidia exogenously

149. The scientist who called the extract of infected tobacco plant as *Contagium vivum fluidum* was

(1) D.J. Ivanowsky

(2) Pasteur

(3) W.M. Stanley

(4) M.W. Beijerinek

150. *Mycoplasma*

(a) Lack cell wall

(b) Can survive without oxygen

(c) Are primitive organisms responsible for production of biogas

(d) Are smallest living organisms

(1) Only (a) & (b) are correct

(2) Only (a) & (d) are correct

(3) Only (c) is incorrect

(4) Only (a) & (d) are incorrect

ZOOLOGY

SECTION-A

151. In which type of epithelial tissue, are the nuclei located closer to the base of cells?

(1) Simple columnar epithelium

(2) Simple cuboidal epithelium

(3) Simple squamous epithelium

(4) Compound cuboidal epithelium

152. Match column-I with column-II and choose the option containing only **correct** match.

	Column-I		Column-II
A.	Tight junctions	p	Perform cementing to keep neighbouring cells together
B.	Adhering junctions	q	Help to stop substances from leaking across a tissue
C.	Gap junctions	r	Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells

(1) A(p), B(q), C(r)

(2) A(q), B(r), C(p)

(3) A(r), B(p), C(q)

(4) A(q), B(p), C(r)

153. All of the following are true for cartilage **except** that

(1) Its matrix is solid and pliable

(2) It is dense irregular connective tissue

(3) It is considered as skeletal connective tissue

(4) Its matrix is generally devoid of inorganic salts

154. Which one exerts the greatest control over the body's responsiveness to changing conditions?

(1) Epithelial tissue

(2) Muscular tissue

(3) Neural tissue

(4) Connective tissue

155. Which of the following is correct w.r.t the properties of areolar and adipose tissues?

(1) Both are examples of dense connective tissue

(2) Both are present beneath the skin

(3) Both of them do not store fats

(4) Both of them show compact packing of fibres and fibroblasts

156. The biomolecule that **cannot** be hydrolysed further is

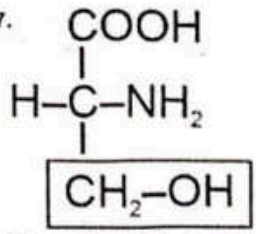
(1) Starch

(2) Insulin

(3) Lactose

(4) Glucose

157.



The amino acid represented by the structure given above is

(1) Glycine

(2) Alanine

(3) Serine

(4) Valine

158. Ester bond in nucleic acid involves

(1) OH of sugar and nitrogen of nitrogenous base

(2) Phosphate and hydroxyl group of sugars

(3) Hydroxyl groups of two sugars

(4) Phosphate and hydroxyl group of nitrogenous base

159. Match column I (Amino acid) with column II (Chemical nature) and choose the **correct** option.

	Column I (Amino acid)		Column II (Chemical nature)
a.	Glutamic acid	(i)	Aromatic
b.	Lysine	(ii)	Neutral
c.	Valine	(iii)	Basic
d.	Tyrosine	(iv)	Acidic

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

160. Glucosamine and N-acetyl glucosamine are chemically modified

(1) Nucleosides

(2) Sugars

(3) Lipids

(4) Nucleotides

161. Association of co-enzyme with apoenzyme is usually

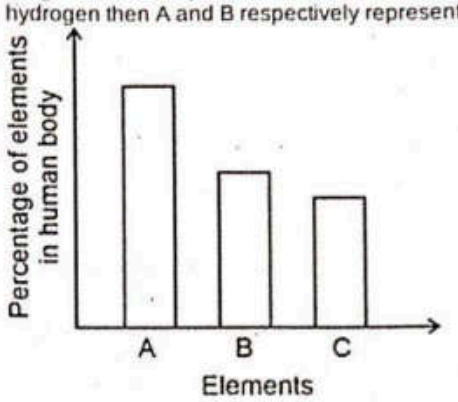
(1) Permanent and loose

(2) Transient and tight

(3) Transient and loose

(4) Permanent and tight

162. The histogram given below represents the percentage weight of element present in human body. If C represents hydrogen then A and B respectively represent



(1) Carbon, Oxygen

(2) Oxygen, Carbon

(3) Sulphur, Carbon

(4) Silicon, Oxygen

163. Phosphorylated organic compound associated with the plasma membrane and present in the macromolecular fraction is

(1) GLUT-4

(2) Lecithin

(3) Guanine

(4) Lysine

164. Molecule 'X' contains sugar attached to nitrogenous base. Phosphate group is also esterified to the sugar. What is molecule X? Choose the **correct** set of 'X' and molecules belonging to this category.

(1) Nucleoside : Adenosine, uridine

(2) Nucleotide : Uridylic acid, cytidylic acid

(3) Nucleic acid : Guanylic acid, adenylic acid

(4) Phospholipid : Lecithin, triglycerides

165. In all of the given carbon compounds, heterocyclic rings can be found, **except**

(1) Cytosine

(2) Cysteine

(3) Cytidine

(4) Uridine

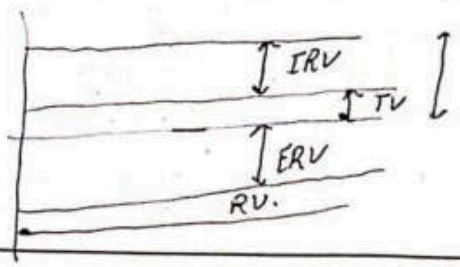
166. Total Lung Capacity (TLC) is **not** denoted by

(1) TLC = VC + RV

(2) TLC = IC + ERV

(3) TLC = IC + FRC

(4) TLC = TV + IRV + ERV + RV



167. Choose the **incorrect** statement w.r.t. respiratory organs/systems in different animals.

(1) Most of the aquatic arthropods and molluscs use gills for respiration.

(2) Tracheal tube system is characteristic of insects.

(3) Amphibians respire through moist skin only in aquatic habitat.

(4) During respiration, fishes use gills while reptiles use lungs.

168. Which of the following structures in humans is not supported by C-shaped cartilaginous rings to prevent from collapsing?

(1) Trachea

(2) Tertiary and secondary bronchi

(3) Primary bronchi

(4) Terminal bronchiole

169. 500 mL of oxygenated blood can deliver how much O_2 to the tissues under normal physiological conditions in an adult human?

(1) 100 mL

(2) 50 mL

(3) 25 mL

(4) 125 mL

170. A person tries to hold his breath for one minute. Which of the following changes in blood would first lead to an urge to breathe?

(1) Fall in O_2 concentration

(2) Rise in CO_2 concentration

(3) Rise in pH

(4) Fall in osmolarity

171. Diffusion membrane in lungs comprises

(1) Thin columnar epithelium of alveoli, endothelium of pulmonary capillaries and basement membrane

(2) Ciliated squamous epithelium of alveoli, endothelium of pulmonary capillaries and basement substance between two

(3) Thin squamous epithelium of alveoli, endothelium of alveolar blood capillaries and basement substance between them

(4) Ciliated columnar epithelium of alveoli, endothelium of alveolar capillaries and basement membrane between them

172.

$$CO_2 + H_2O \overset{\text{Carbonic anhydrase}}{\rightleftharpoons} H_2CO_3 \overset{\text{Carbonic anhydrase}}{\rightleftharpoons} HCO_3^- + H^+$$

The above reaction occurs in the backward direction at the

(1) Tissue site where pCO_2 is high

(2) Alveolar site where pCO_2 is low

(3) Tissue site where pO_2 is high

(4) Alveolar site where pO_2 is low

173. Factor **not** favourable for formation of oxyhaemoglobin is

(1) High pO_2

(2) Low pCO_2

(3) Lesser H^+ concentration → Low pCO_2

(4) High temperature

174. In humans, approximate percentage of CO_2 transported to lungs from tissue through blood plasma is

(1) 23%

(2) 7%

(3) 77%

(4) 30%

175. Match the following columns and select the **correct** option.

	Column I		Column II
a.	Heart attack	(i)	Blood is not pumped effectively enough to meet the needs of body
b.	Heart failure	(ii)	Heart stops beating
c.	Cardiac arrest	(iii)	Heart muscles are suddenly damaged due to inadequate blood supply

(1) a(ii), b(iii), c(i)

(2) a(iii), b(i), c(ii)

(3) a(i), b(ii), c(iii)

(4) a(iii), b(ii), c(i)

176. Megakaryocytes on fragmentation give rise to

(1) RBCs

(2) WBCs

(3) Platelets

(4) Macrophages

177. Most abundant nucleated granulocytes in blood of a healthy adult man are

(1) Erythrocytes

(2) Lymphocytes

(3) Neutrophils

(4) Eosinophils

178. In a standard ECG, QRS complex represents an event in which there is

(1) Production of second heart sound characterised by low pitch

(2) Rise in intraventricular pressure

(3) Closure of semilunar valve

(4) Flow of blood into the atria from ventricles

179. Pulmonary veins from lungs are known for carrying A blood to the B of the heart. Choose the option that fills A and B **correctly**.

	A	B
(1)	Oxygenated	Right atrium
(2)	Deoxygenated	Right ventricle
(3)	Oxygenated	Left atrium
(4)	Deoxygenated	Left ventricle

(1) (1)

(2) (2)

(3) (3)

(4) (4)

180. Which type of blood contains anti B-antibodies in the plasma and lacks 'Rh' group antigens?

(1) 'AB' negative

(2) 'A' negative

(3) 'B' negative

(4) 'O' positive

181. Erythroblastosis foetalis can be prevented by administering anti-Rh antibodies to

(1) Rh –ve mother post partum Rh +ve baby

(2) Rh +ve mother post partum Rh +ve baby

(3) Rh –ve mother post partum Rh –ve baby

(4) Rh +ve mother post partum Rh –ve baby

@NEETBLADE4

182. If the sino-atrial node is damaged or diseased, which of the following is most likely to happen?

(1) The heart stops beating

(2) The heart beats at a faster pace

(3) Heart beats at a slower rate

(4) Heart initially beats faster and later the heart beat decreases

183. Serum differs from plasma in

(1) Absence of clotting factors

(2) Presence of albumin proteins

(3) Presence of immunoglobulin

(4) Presence of clotting factors

184. What is the normal range of haemoglobin per 100 mL of blood in a healthy adult human?

(1) 20 - 25 gm

(2) 5 - 6 gm

(3) 12 - 16 gm

(4) 72 - 80 gm

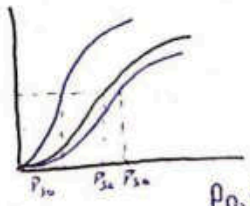
185. The factor that shifts oxygen–haemoglobin dissociation curve to the left side is

(1) High partial pressure of oxygen

(2) High partial pressure of carbon dioxide

(3) High body temperature

(4) High H^+ concentration



SECTION-B

186. A type of dense regular connective tissue, which connects skeletal muscles to bone is

(1) Ligament

(2) Cartilage

(3) Tendon

(4) Fat body

187. The fibres of which of the following types of muscular tissues are multinucleated and show alternate light and dark bands?

(1) Smooth muscle tissue

(2) Skeletal muscle tissue

(3) Cardiac muscle tissue

(4) Visceral muscle tissue

188. Read the following statements and choose the **correct** option.

Statement A : The structure of a cell varies according to its function.

Statement B : Depending on location, structure and function, the animal tissues have been broadly classified into four general types.

(1) Both statements A and B are incorrect

(2) Only statement A is correct

(3) Only statement B is correct

(4) Both statements A and B are correct

TEAM NEET BLADE

189. Inulin is a polymer of

(1) Glucose

(2) Fructose

(3) Sucrose

(4) Cellulose

190. Enzyme pepsin works in the pH range 1.5 to 2.5. What would happen to the rate of the reaction controlled by this enzyme, if there is a rise in pH from 2.5 to 8?

(1) The rate doubles

(2) The rate starts to decrease

(3) There will be no effect on rate

(4) The rate first increases then decreases

191. During conversion of substrate to product, transition states formed are

(1) Stable and transient

(2) Temporary and unstable

(3) Low energy states

(4) Permanent altered structural states

192. With the increase in substrate concentration, the velocity of the enzymatic reaction rises and ultimately reaches a maximum velocity. What happens to the enzymatic activity when more substrate is added in this condition?

(1) It will decrease

(2) It will increase

(3) It will remain unchanged

(4) It will first increase then it will decrease

193. A person starts coughing suddenly while swallowing some food. It could be due to improper movement of

(1) Neck

(2) Diaphragm

(3) Tongue

(4) Epiglottis

194. In humans, inspiration is possible if the pressure within the lungs is

(1) Less than the atmospheric pressure

(2) More than the atmospheric pressure

(3) Equal to the atmospheric pressure

(4) Double than the atmospheric pressure

195. Read the following given statements and choose the **correct** option.
Statement-A: Cutaneous respiration occurs through vascularised bags, called lungs.
Statement-B: Pulmonary respiration occurs through the moist skin.

(1) Only statement A is incorrect

(2) Only statement B is incorrect

(3) Both statements A and B are incorrect

(4) Both statements A and B are correct

196. Match column I and column II w.r.t respiratory volumes in mL of an adult human under normal conditions.

	Column I		Column II
a.	Tidal volume	(i)	1100–1200
b.	Inspiratory reserve volume	(ii)	1000–1100
c.	Expiratory reserve volume	(iii)	2500–3000
d.	Residual volume	(iv)	500

Choose the **correct** option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

197. The nodal tissue located in the lower left corner of right atrium close to inter-atrial septum is

(1) SA node

(2) AV node

(3) AV bundle

(4) Purkinje fibres

198. Tunica media is thickest in the wall of

(1) Vein

(2) Capillary

(3) Artery

(4) Lymph vessel

199. How many cardiac cycles are completed in a normal healthy person per minute?

(1) 16

(2) 72

(3) 0.8

(4) 4.0

200. **Assertion :** Dub is the second heart sound.
Reason : It is associated with the closure of the atrio-ventricular valves.
In the light of above statements, select the **correct** option.

(1) Both Assertion & Reason are true and Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.